

1) Hidden Layer Linear Combination

First Neuron) $z_1 = W_1 x + b_1$

$$z_{11} = (0.1)(0.5) + (0.2)(0.1) \\ = 0.05 + 0.02 = 0.07$$

Second Neuron

$$z_{12} = (0.3)(0.5) + (0.4)(0.1) \\ = 0.15 + 0.04 = 0.19$$

Therefore

$$z_1 = \begin{bmatrix} 0.07 \\ 0.19 \end{bmatrix}$$

2) Hidden Layer Activation

Apply sigmoid each value

First Neuron:

$$a_{11} = \sigma(0.07)$$

$$= \frac{1}{1 + e^{-0.07}}$$

$$e^{-0.07} = 0.9324$$

$$a_{11} = \frac{1}{1 + 0.9324} = 0.5175$$

Second Neuron

$$a_{12} = \sigma(0.19)$$

$$= \frac{1}{1 + e^{-0.19}}$$

$$e^{-0.19} = 0.8270$$

$$= \frac{1}{1 + 0.8270} = 0.5474$$

Therefore $a_1 = \begin{bmatrix} 0.5175 \\ 0.5474 \end{bmatrix}$

3) Output Layer Linear combination

$$z_2 = w_2 a_1 + b_2$$

$$= (0.5)(0.5175) + (0.6)(0.5474)$$

$$= 0.2588 + 0.3284$$

$$z_2 = 0.5872$$

4) Output Prediction

$$\hat{y} = \sigma(z_2)$$

$$= \frac{1}{1 + e^{-0.5872}}$$

$$e^{-0.5872} = 0.5558$$

$$\hat{y} = \frac{1}{1 + 0.5558} = 0.6428$$

5) Compute the Loss L

$$L = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$

since

$$y = 1$$

$$L = -(1) \log(0.6428) - (1-1) \log(1-0.6428)$$

Simplify

$$1-1=0$$

So

$$0 \times \log(1-0.6428) = 0$$

Therefore

$$L = -\log(0.6428)$$

using calculator

$$\ln(0.6428) \approx -0.4420$$

$$L = -(-0.4420)$$

$$\boxed{L = 0.4420}$$